

Project Files

Python Scripts:

- `constraint_moving_helipad.py`: Lunar Lander with moving landing pad
- `constraint_random_force.py`: Lunar Lander with random initial force
- `constraint_random_inicial_x.py`: Lunar Lander with random initial X position
- `constraint_wind.py`: Lunar Lander with wind constraints
- `soft_precision_landing_no_wind.py`: Environment in which we trained our best agent (the constraints will be applied in this agent)
- `evaluation_of_constraints.py`: Script to evaluate trained agents
- `train.py`: Training script for PPO agents
- `visual.py`: Visualization script to display trained agent in action

Additional Folders:

- `logs/`: Training logs and TensorBoard data
- `models/`: Pre-trained agent models

Documentation:

- `README.pdf`: This file
- `presentation_slides.pdf`: Presentation slides for the project

Folder Structure

```
models/
├── Baseline
│   ├── ppo_baseline_1m.zip
│   └── ppo_baseline_500k.zip
├── Baseline+CNN
├── Feet
├── HD+Penalties
├── HorizontalDescent
├── MaxFuel
├── PrecisionLanding
├── Soft+Precision
├── Soft+Precision+1M
│   ├── ppo_soft_precision_landing_1m.zip
│   └── ppo_soft_precision_no_wind.zip
├── Soft+Precision+Bandeiras
├── Soft+Precision+Bandeiras+1M
├── Soft+Precision+CNN
└── SoftLanding
```

Note: Folders with multiple items are shown with their contents listed.
Other directories contain trained model files (PPO agents in .zip format)
and do not have multiple files within them.

Description of Model Folders

Baseline

- Standard Lunar Lander agent without constraints

Baseline+CNN

- Baseline agent using CNN-based architecture

Feet

- Customized agent with the addition of feet.

HD+Penalties

- Horizontal Descent agent with penalties

HorizontalDescent

- Agent trained for horizontal descent strategy

PrecisionLanding

- Agent trained for precise landing positions

Soft+Precision

- Agent combining soft landing and precision landing rewards

Soft+Precision+1M

- Soft+Precision agent trained for 1 million steps

Soft+Precision+Bandeiras

- Soft+Precision agent with flag landing zones

Soft+Precision+Bandeiras+1M

- Soft+Precision+Bandeiras agent trained for 1 million steps

Soft+Precision+CNN

- Soft+Precision agent using CNN-based architecture

SoftLanding

- Agent trained for soft landing with low impact velocity

Corresponding Training Logs

Each trained model in the models/ folder has a corresponding training log directory in the logs/ folder. These logs contain TensorBoard data that can be used to visualize training metrics and performance graphs for each agent.

To view training graphs for a specific model using TensorBoard:

1. Open terminal/command prompt
2. Navigate to the project directory
3. Run: `tensorboard --logdir=logs`
4. Open the URL displayed in your browser (typically `http://localhost:6006`)